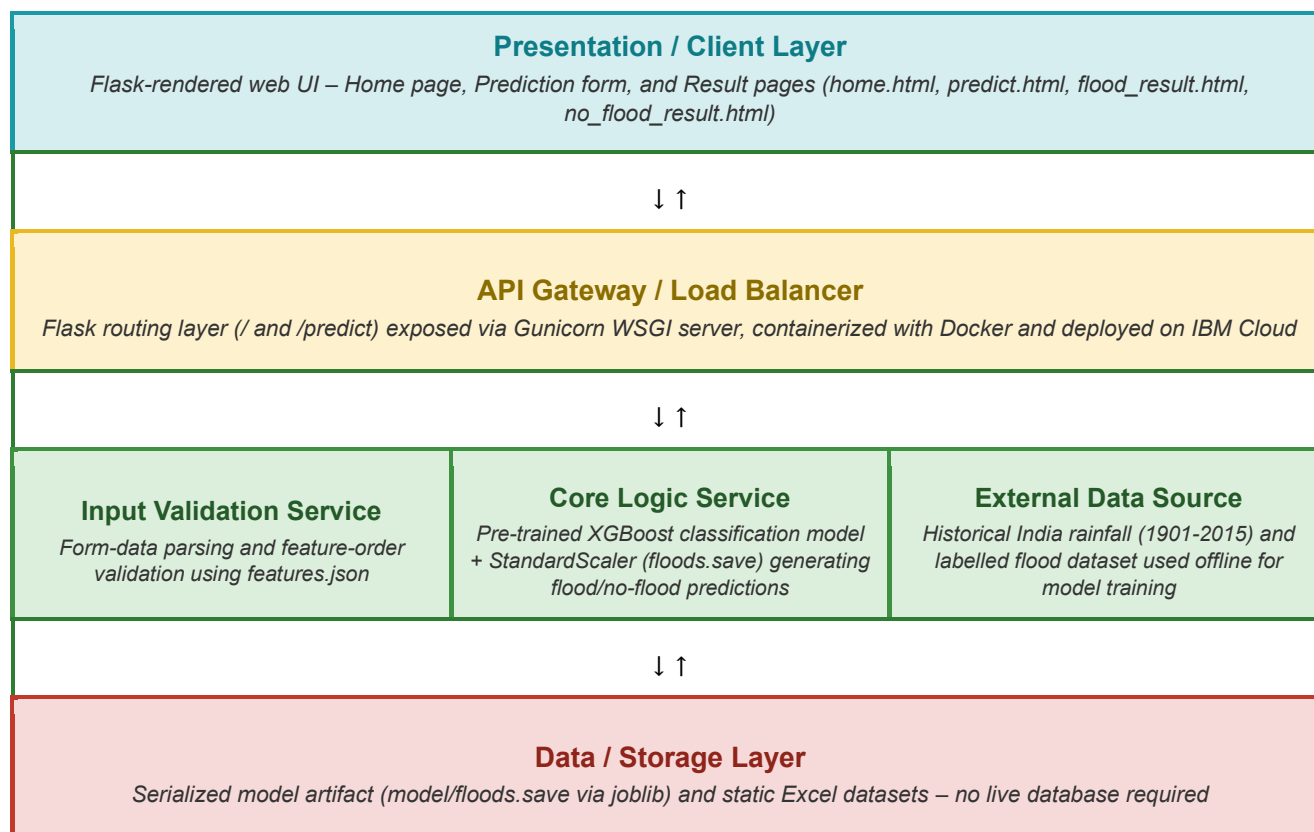


Solution Architecture

Date	04 July 2026
Team ID	
Project Name	Rising Waters - ML-Based Flood Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level view of the Rising Waters architecture, showing the Presentation Layer, Application Layer, and Data Layer used to serve real-time flood predictions.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the home page, the prediction input form, and the flood / no-flood result pages returned to the user.	Flask (Jinja2 templates: home.html, predict.html, flood_result.html, no_flood_result.html), HTML, CSS
API Gateway	Handles incoming HTTP requests, routes GET requests to the form page and POST requests to the prediction endpoint.	Flask routing (/ and /predict), Gunicorn WSGI server, Docker, IBM Cloud
Core Logic / ML Service	Loads the trained model bundle, scales incoming feature values, and generates the flood-risk prediction and probability.	Python, XGBoost, scikit-learn (StandardScaler), joblib, pandas, numpy
Data / Storage	Stores the serialized model artifact and the historical datasets used to train and validate the model; no runtime database is used.	joblib (model/floods.save), Excel (.xlsx) datasets, local/container file system